Prince Agyei Tuffour

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SUMMARY

Software engineer with a robust background in QA, development, and machine learning, skilled in Python, Java, and JavaScript. Demonstrated expertise in enhancing product reliability by implementing advanced testing frameworks and driving user engagement through scalable, mobile-first platforms. Proven leadership in managing full-cycle development, integrating ML solutions, and optimizing systems for high-performance operations.

SKILLS

- **Languages**: Python, Javascript, C++, SQL, Java, Typescript
- **Tools & Technologies**: Git, Nextjs, Redis, AWS, Google Cloud, Playwright, Tailwind CSS, Keras, Pandas, ScikitLearn, Matplotlib, Deep Learning, Linux/Unix, PyTorch, Tensorflow, Selenium, Flask, Django, RESTful APIs, Express, PostgreSQL, MongoDB, React, SciPy, Streamlit, YOLO

WORK EXPERIENCE

dynaConnections Corporations **Mar 2025 - Present**

Software QA Engineer

- Designed and executed comprehensive test strategies employing Python and JavaScript for manual and automated testing, API validation, user interface checks, and performance testing to ensure industry compliance.
- Achieved a 40% improvement in overall product reliability and a 60% reduction in post-deployment bugs by implementing quality assurance best practices and establishing testing frameworks that detected edge cases early.
- Developed detailed test cases and conducted regression testing while applying computer science fundamentals and analytical problem-solving to enhance the software development lifecycle and user experience.

Akomapa Health **Mar 2025 - Present**

Lead Software Developer/Executive Board Member

- Managed complete website workflow including server maintenance and database optimization using AWS cloud services, achieving 99.8% uptime reliability for a health education platform serving over 500 active monthly users.
- Developed automated content delivery systems and implemented mobile-first design improvements using JavaScript, HTML, and CSS3 principles, increasing user engagement by 65% and boosting mobile traffic by 80%.
- Built an interactive community platform enabling students to access peer-reviewed health content, participate in wellness challenges, and connect with certified health professionals, demonstrating technical leadership and strategic problem-solving.

Cita Marketplace.com **Jun 2022 - 2023**

Machine Learning Engineer - Intern

- Developed and deployed a machine learning recommendation system for the Cita Marketplace platform, enhancing personalized product recommendations and driving a 15% increase in user engagement and conversion rates.
- Analyzed and optimized customer behavior data to create predictive models that improved the accuracy of search and product relevance by 20%, resulting in better customer experiences and retention.
- Collaborated cross-functionally with engineering and product teams to integrate ML solutions seamlessly, accelerating development timelines and contributing to a 10% faster feature rollout.

Oregon State University **Sep 2021 - Dec 2023**

Graduate Research Assistant

- Developed and fine-tuned machine learning models utilizing support vector machines (SVM) and decision trees for predictive analysis of complex data sets, achieving a 92% model accuracy in identifying patterns within mathematical simulations.
- Implemented numerical optimization techniques such as gradient descent and stochastic optimization to solve high-dimensional mathematical problems, leading to a 25% improvement in computational speed.
- Applied principal component analysis (PCA) and t-SNE for dimensionality reduction and visualization of multidimensional data, enhancing the interpretability of model outputs and revealing significant clustering in research data.
- Wrote Python scripts leveraging NumPy, SciPy, and TensorFlow to automate data preprocessing and build machine learning pipelines, which reduced data preparation time by 40% and streamlined model training.

EDUCATION

Oregon State University, Corvallis OR

Master of Science, Mathematics

Kwame Nkrumah University of Science and Technology, Ghana

Bachelor of Science, Mathematics

PROJECTS

GradGPT.pro

- Designed and launched GradGPT.pro, an AI-powered graduate school assistant using GPT-4, streamlining school selection, application prep, and funding search for 1,500+ users.
- Integrated scraping tools, LLM-based Q&A, and secure chat saving to deliver tailored insights, boosting application efficiency and success for early-stage grad applicants.

Akomapa Health
- Engineered a Yale- and UCLA-backed health platform using React, Supabase, and TypeScript, reaching 25,000+ users and improving health literacy by 65% in rural Ghana.
- Achieved 80% rise in preventive care in high blood pressure and 300% outreach growth via offline features, volunteer medical training, and integrated emergency contact systems.

ByDay Jobs - Demo App
- Designed and built a location-based service platform (like Uber for professionals) using React, Next.js, Google Maps API, and Stripe to connect skilled workers with local demand in real-time.
- Currently in pilot testing with early users across 5 cities; features smart job matching, secure payments, and user reviews, laying the groundwork for scalable job creation and gig economy growth in underserved markets.

Stay Afloat - Customizable LMS for Ghanaian Institutions
- Developed a flexible SaaS learning platform tailored for Ghanaian institutions, enabling remote students to access quality online education with customizable branding and course structures.
- Empowered 12+ schools to expand online reach, achieving 60% higher course completion rates and 200% increase in student engagement through localized features, payment integration, and real-time progress tracking.

GPU Memory Profiler for TensorFlow & PyTorch - Open Source
- Contributed to GPU memory profiling tool optimizing deep learning workflows by visualizing model memory usage for TensorFlow and PyTorch.
- Enhanced debugging, training efficiency, and memory usage visualization by identifying bottlenecks, aiding various ML practitioners in improving model performance.